Project Report: Ames Housing Data Analysis

Prepared by: Mohannad Alrajhi

Introduction

In this project, I used the Ames Housing Dataset, which contains detailed information about residential properties sold.

The main questions I aimed to answer were:

- What factors affect house prices?
- How different features relate to the target (SalePrice)?
- How to prepare the data for machine learning?

Cleaning summary

During data exploration, I found several issues:

Missing Values

Some columns had missing values such as:

- Garage-related features
- Basement features
- Electrical

How I fixed them:

- For categorical columns: I filled missing values with "None"
- For numerical columns with very few missing values: I filled them with the median
- Some values were logically missing (e.g., no garage → no garage data)

Outliers

I checked outliers using a boxplot on the target column (SalePrice).

What I did:

- I capped extreme values at the 99th percentile to reduce their impact

Feature engineering summary

To gain deeper insights, I created new informative columns:

- **Age Group:** I transformed the Year Built column into categories (New, Mid-Age, Old). This helped in comparing price trends across different eras.

- **House Age:** I created a house_age feature by subtracting Year Built from Yr Sold. I then categorized this into (New, Mid-Age, Old) to better analyze how depreciation affects market value.

Key findings

1. The Power of Quality: I found that houses with a quality rating of 8 or 9 are much more expensive than others, showing that materials and finish quality are "deal breakers" in the market.

---

2. Size Really Matters:

---

3. Newer Houses Win

New houses (0-10 years) have the highest average prices, while old houses are the cheapest, proving that age reduces property value.

Best chart

Next step

If I had more time, I would:

- Build and compare multiple models (Linear Regression, Random Forest).
- Perform feature selection more carefully.
- Tune hyperparameters to improve accuracy.
- Fix all wrong type.
- Visualize more relationships between features.

Conclusion

This project helped me understand:

- Data cleaning
- Feature engineering
- Data transformation
- Basic analysis